

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

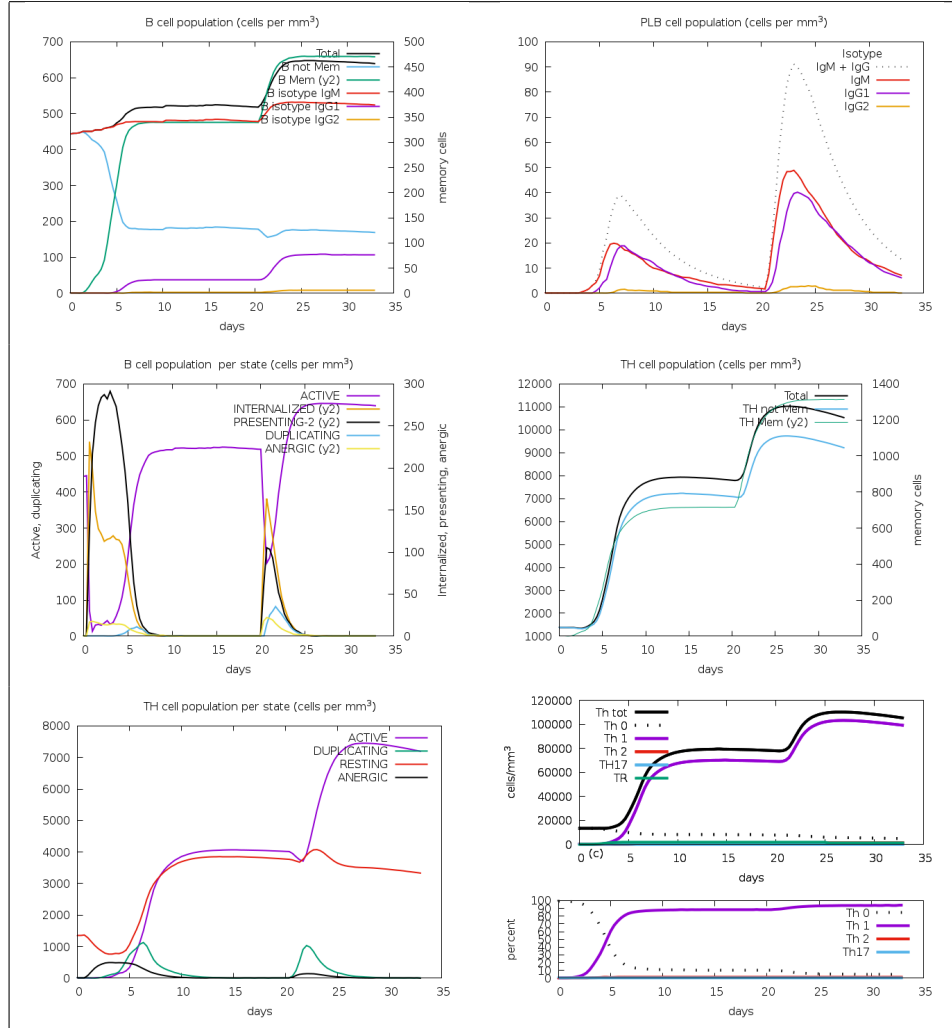


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

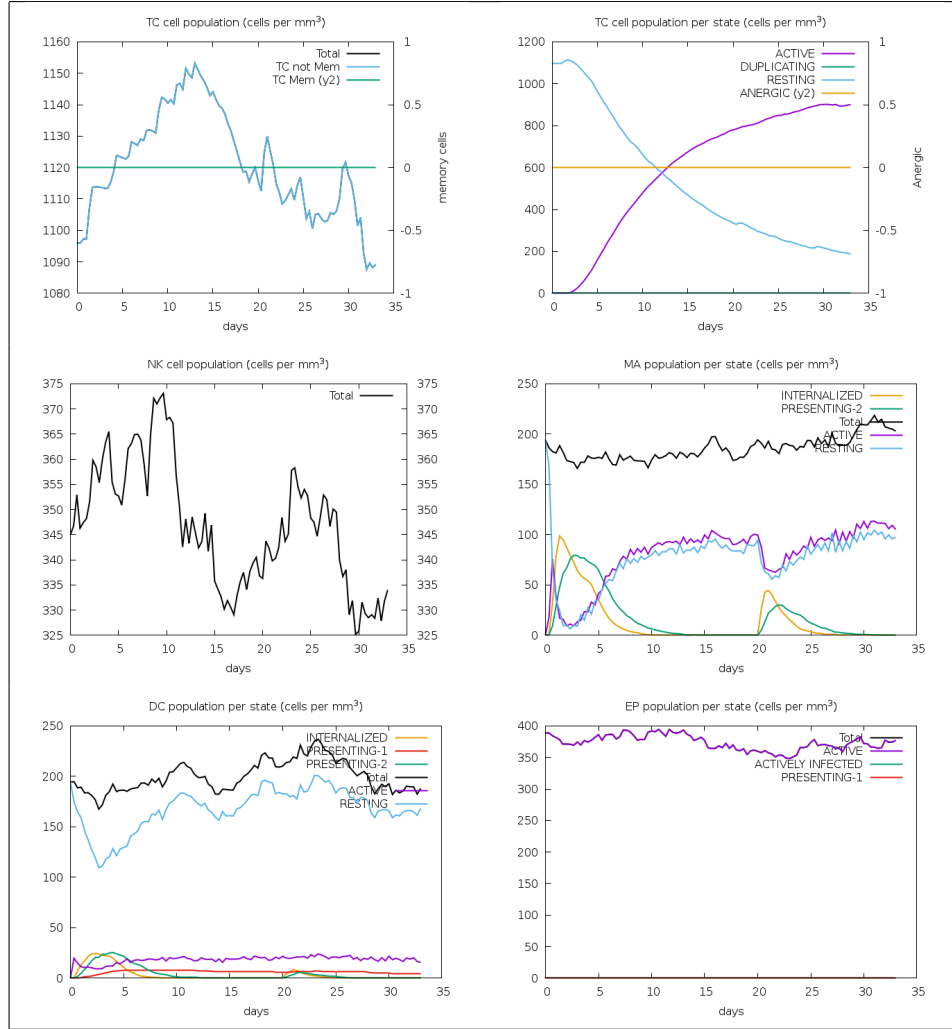


Figure 2: Legend: symbols as figure above.

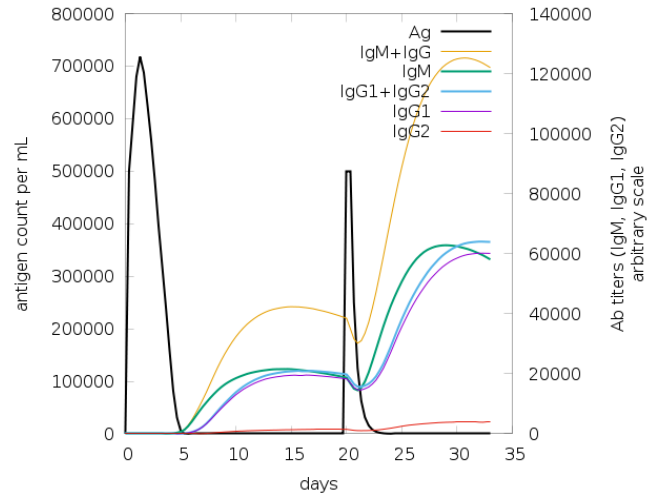


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

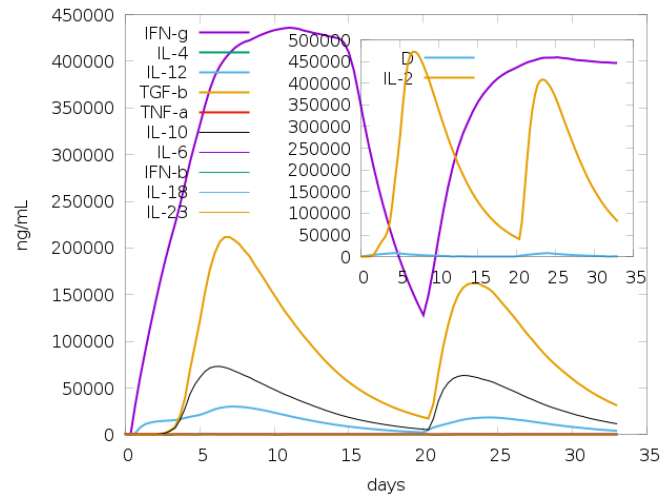


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Use Parker's propensity scale, takes an antigen block as input, and creates a list of residues that are possible epitopes.

[illegible][illegible]

```

1]      pos=12 len=10      AGGSGPGTGP
2]      pos=25 len=53      LGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRH
3]      pos=79 len=4       DGVR
4]      pos=95 len=66      GTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAGGGAGAGGGAGGAG
5]      pos=207 len=7      TEKGPGP

```

Given the antigen injected creates the list of peptides for all the NumAgProts proteins and for all i.e., 4 MHC I molecules

Read class I peptide list from file? NO

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

AKFVAAWTLKAAAGGSGPGTGPNGNLGEKGDTSGPEGGSGGSPQRRGGDNHGRGRGRGRGGGRGPAPGGSGSGPRHRD
GVRRPQKRPSCIGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAGGGAGGAGAGGGA
GAEAAAKFLDKGTYTLGPGPGTVMIEAGLYGPGPGTSKFLMGTYKRVTEKGPGRHLLVSGAEAAAKLPWPFPMPV

```
0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event
```

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013404_5_tJixyoRfmyzL.FSA_1_001

AKFVAAWTLKAAAGSGPGTGPNGLGEKGDTSPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGGA
GAEEAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGRHLLVSGAEAAKLPWFPPMV

Epitopes of protein 0 -----

0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event

Allele: B0702

Pseudo sequence: KAAREEQIKAQTRE

Threshold: 8.702800

Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013404_5_tJixyoRfmyzL.FSA_1_001

AKFVAAWTLKAAAGSGPGTGPNGLGEKGDTSPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGGA
GAEEAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGRHLLVSGAEAAKLPWFPPMV

Epitopes of protein 0 -----

0] pos= 64 score=0.002728 unnormalised=0.3012000000 RPGAPGGSG
1] pos= 83 score=0.017497 unnormalised=1.9322000000 RPQKRPSCI
2] pos= 190 score=0.004023 unnormalised=0.4442000000 GPGPGTSKF
3] pos= 210 score=0.026354 unnormalised=2.9102000000 GPGGRHLL
4] pos= -1 score=0.949399 unnormalised=104.8410000000 non-binding event

Allele: B0702

Pseudo sequence: KAAREEQIKAQTRE

Threshold: 8.702800

Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013404_5_tJixyoRfmyzL.FSA_1_001

AKFVAAWTLKAAAGSGPGTGPNGLGEKGDTSPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGAGAGGGGA
GAEEAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGRHLLVSGAEAAKLPWFPPMV

Epitopes of protein 0 -----

0] pos= 64 score=0.002728 unnormalised=0.3012000000 RPGAPGGSG
1] pos= 83 score=0.017497 unnormalised=1.9322000000 RPQKRPSCI
2] pos= 190 score=0.004023 unnormalised=0.4442000000 GPGPGTSKF
3] pos= 210 score=0.026354 unnormalised=2.9102000000 GPGGRHLL
4] pos= -1 score=0.949399 unnormalised=104.8410000000 non-binding event

DoPeptideList_II:

Given the antigen injected creates the list of peptides for all the
NumAgProts proteins and for all i.e., 2 MHCII molecules

Read class II peptide list from file? NO

=====
Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013404_5_tJ1xyoRfmyzL.FSA_1_001

AKFVAAWTLKAAAGSGPGTGPNGLGKGDTSPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSICIGCKGTHGGTGAGAGAGGAGAGAGGGAGAGGGAGGAGAGGAGAGGGAGAGGGAGGAGAGGGGA
GAEEAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGRHLLVSGAEAAKLPWFPPMV

Epitopes of protein 0 -----

0]	pos=	2	score=0.040812	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.005566	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	8	score=0.040800	unnormalised=3.3835600000	LKAAAGGSG
3]	pos=	173	score=0.077469	unnormalised=6.4245600000	YTLGPGPGT
4]	pos=	198	score=0.011366	unnormalised=0.9425600000	FLMGTYKRV
5]	pos=	218	score=0.026113	unnormalised=2.1655600000	LLVSGAEAA
6]	pos=	-1	score=0.797874	unnormalised=66.1680000000	non-binding event

=====
Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013404_5_tJ1xyoRfmyzL.FSA_1_001

AKFVAAWTLKAAAGSGPGTGPNGLGKGDTSPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGGSGSGPRHRD
GVRRPQKRPSICIGCKGTHGGTGAGAGAGGAGAGAGGGAGAGGGAGGAGAGGGAGAGGGAGGAGAGGGGA
GAEEAAKFLDKGTYTLGPGPGTVMEIAGLYGPGPGTSKFLMGTYKRVTEKGPGRHLLVSGAEAAKLPWFPPMV

Epitopes of protein 0 -----

0]	pos=	2	score=0.040812	unnormalised=3.3845600000	FVAAWTLKA
1]	pos=	6	score=0.005566	unnormalised=0.4615600000	WTLKAAAGG
2]	pos=	8	score=0.040800	unnormalised=3.3835600000	LKAAAGGSG
3]	pos=	173	score=0.077469	unnormalised=6.4245600000	YTLGPGPGT
4]	pos=	198	score=0.011366	unnormalised=0.9425600000	FLMGTYKRV
5]	pos=	218	score=0.026113	unnormalised=2.1655600000	LLVSGAEAA
6]	pos=	-1	score=0.797874	unnormalised=66.1680000000	non-binding event